Anan Faiyaz Radin

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Objective: *A driven UoN MSc CS(AI) student with exceptional research and teaching acumen looking for graduate research opportunities in the areas of Robotics, Dynamical Control Systems, Computer Vision, Machine Learning & Biomedical Engineering.*

Education

- **MSc in Computer Science (AI),** University of Nottingham, UK
  - 2nd Term (Ongoing) — 2025 - Present
- **BSc in Electrical & Electronic Engineering,** North South University, BD
  - CGPA: **3.53**/ 4.0 (Cum Laude)
    Focus: Robotics, VLSI Design — 2019 - 2024

Research Experience

Quantum Mechanical Trapping of a Spinless Particle using Matterwaves — 2024 - 2025
Supervisor: Dr. Mahdy Rahman Chowdhury (Research Training)

- Replicated published results of previous work in COMSOL Multiphysics and Quantum Espresso; And, attempted a novel approach to quantum mechanical trapping of a spinless particle using matter waves.

Pose Estimation of Target Spacecraft from a 3U CubeSat with Deep Learning Methods — 2024
Supervisor: Dr. Shahnewaz Siddique (APSCO Competition & Undergraduate Thesis)

- Derived mission parameters for stable pose estimation of target spacecraft in Low Earth Orbit, developed an Attitude Determination & Control System model in MATLAB for the said parameters.

Professional Experience

Emerging Leader, Edward M. Kennedy Center, American Spaces Bangladesh — 2025

- Hosted hands-on robotics workshops in the EMK Makerlab for school, and university students. Also, processed & delivered 3D Printing projects using Prusa and SOLIDWORKS.

Research Assistant, Department of ECE, North South University — 2024 - 2025

- Designed and implemented a Quaternion-based Linear Quadratic Regulated attitude controller for a 3U CubeSat in MATLAB under SNS1.

Teaching Assistant, Department of ECE, North South University — 2024 - 2024

- Evaluated student submissions, & co-ordinated course instruction for EEE413: Verilog HDL under IQR for a peer group of 40.

Instructor (Robotics), Mindcraft Labs — 2019-2020

- Taught an introductory Arduino robotics curriculum through practical, project-based sessions.

Student Intern (R&D), Bondstein Technologies Ltd. — 2016 - 2019

- Designed system process for a automated car parking project, PCB layout for Sensor Arrays, & CAD modeling for R&D project housings.

Technical Skills

- **Programming/Dev** • LaTeX • Python[2025] • C++ • HTML/CSS [2025] • PHP [2025] • JavaScript [2025] • Github[2025] • Verilog HDL • Arduino • MPLAB X IDE • Assembly [Familiar]
- **Data Analytics** • MS Excel • Tableau [2024] • Pandas [2025]
- **Simulation Tools** • MATLAB • Cadence • COMSOL • AutoCAD • MULTISIM • Solidworks • EasyEDA • Xilinx • Wolfram Mathematica • Manim [Familiar] • ROS [Familiar]
- **Graphics** • Adobe Creative Suite (Illustrator, Photoshop, After Effects, Premiere Pro, XD)
- **Certifications** • Stanford Short Data Science Course (MS Excel, SQL, AI, Tableau)

Standardized Tests

- **GRE** • Quant: **160** • Verbal: 149 • Analytic Writing: 4.0 — 2024
- **IELTS,** IDP • Band Score: **8.0**/ 9 — 2024
- **SAT,** College Board • Score: **1480**/ 1600 • 98th Percentile (99th Nationally) — 2018

Achievements & Awards

- **Finalist & Representative Team for Bangladesh**, APSCO CubeSAT Competition — 2024
- **Champions** (Out of 23 Teams), Intra Masters of Ideation — 2021
- **Full Tuition Merit UG Scholarship** ($35,000/year), School of Engineering, University of Arizona, USA — 2019
- **Finalist** (Top 10), Engenius Line Following Robotics Competition, Ahasanullah University of Technology — 2017

Leadership Experience

President (27th), North South University Young Entrepreneurs Society! (NSU YES!), NSU — 2019-2023

- As the 1st STEM background appointee in the organization's history, I led major structural, and operational improvements. I developed a new HR Tracker, organized the Silver Jubilee for 500+ alumni, & led 7+ national-scale youth development events, and co-founded the creative platform Make IT Happen and secured £20,000+ in sponsorship funding. Besides, I supervised, and trained 200+ members.

Founding President, Rajuk College Robotics Club (RCRC), RUMC — 2015-2019

- Founded the 1st school robotics club in Bangladesh, now known as RC-RIC. Over 4 years, served as moderator and instructor to 400+ students. Designed and executed a 3-year plan 2 inter-school workshops & 8+ national competitions to normalize robotics across Bangladeshi schools.

Academic & Co-curricular Projects

A Technical Introduction to Quantum Computing — 2025
Supervisor: Dr. Gail Hopkins (Systems & Networks Coursework)

- Presented the mathematical foundations and core principles of quantum computing, using LaTeX and Manim to support precise visualization and communication.

Attitude Determination of a Spacecraft using Simple Kalman Filtering in MATLAB — 2024
Supervisor: Dr. Shahnewaz Siddique (Robotics & Numerical Methods)

- Designed and implemented a MATLAB simulation for Detumbling & Nominal Modes with sensor fusion, analyzed & compared pre-filter & post-filter state readings to evaluate system performance.

Verilog HDL Implementation of Conjugate Logic Circuits in Cadence, Xilinx & ModelSim — 2023-2024
Supervisor: Mr. Iqbalur Rahman Rokon (Introduction to VLSI, Verilog HDL Modelling)

- Designed and implemented multiple logic circuits, performing DC and transient analysis on the layouts to ensure accurate functionality and performance.

Heartbeat Monitor Design & Implementation using PIC Microcontroller — 2023
Supervisor: Dr. Shakhawat Hossain (Embedded Systems & Hardware Interface)

- Designed and implemented a heartbeat monitor(physical & simulation) based on peak detection of Photoplethysmography signals using a PIC-Microcontroller with hardware-in the-loop.

PID-Controlled Quadcopter & Line Following Robot using Arduino — 2014 - 2020
Personal & Team Projects for Competitions

- Built quadcopter and robots (multiple iterations), designed and implemented physical circuits & PID controllers, using sensor fusion and gain value optimization in MATLAB.

Reference

- **Swarupa Haque**
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